

SOFTWARE REQUIREMENTS SPECIFICATION

Placement Management System

Campus Recruitment Workflow Platform

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for a role-based Placement Management System designed to coordinate the campus recruitment workflow between students, faculty placement coordinators, the Training and Placement Officer (TPO), and recruiting companies.

The intentions of this document are threefold. First, it serves as a binding contract between the development team and project stakeholders, fixing the scope of the system at baseline. Second, it provides the design and testing teams with an unambiguous specification from which the design document, test plan, and acceptance criteria are derived. Third, it acts as a reference for future maintainers who will need to understand the rationale behind specific behaviours of the system.

The target audience comprises the following:

- The student development team responsible for implementation.
- The academic supervisor and course evaluator who will assess the project against software-engineering process standards.
- Future maintainers and contributors to the system.
- End users — students, faculty, TPO, and recruiters — who require an authoritative description of system behaviour.

1.2 Scope

The Placement Management System (henceforth "the system" or "PMS") is a web-based application that digitises and centralises the campus placement workflow currently coordinated through fragmented channels such as email, spreadsheets, and informal messaging.

The system shall:

- Provide authenticated, role-aware accounts for four distinct user classes: Student, Faculty, TPO, and Company.
 - Support the complete job lifecycle: creation by a company, assignment to specific schools by the TPO, approval or rejection by the faculty of those assigned schools, and discovery by students belonging to those schools.
 - Enforce per-application resume submission, ensuring tailored applications rather than a single global resume.
 - Allow companies to view applicant data, shortlist candidates, and progress them through subsequent recruitment stages.
 - Coordinate online and offline examinations with company-proposed dates, TPO confirmation, faculty availability checks, and venue allocation as appropriate.
 - Support slot-based interview scheduling with both system-assigned and first-come-first-served selection modes, including time-bound activation of online meeting links.
 - Deliver an announcement and notification layer encompassing email and SMS alerts for all critical events.
 - Provide dashboards and reports to the TPO for placement-rate analysis, drive counts, and system performance.
 - Enforce strict role-based access control so that each user sees only data relevant to their role and scope.
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The system shall not:

- Perform academic record verification with any external university ERP at launch.
- Process stipend disbursement, offer-letter dispatch, or payroll-related operations.
- Provide native mobile applications at launch; the responsive web client is the only delivery channel.
- Replace policy-making bodies; it enforces existing institutional rules but does not define new ones.

1.3 Definitions, Acronyms, Abbreviations

Term	Meaning
PMS	Placement Management System — the software described in this document.
TPO	Training and Placement Officer — the central administrator responsible for placement operations across all schools.
Faculty	A faculty member designated as Placement Coordinator for a particular school.
Company	An external recruiting organisation, represented in the system by its registered recruiter user.
Student	An enrolled member of the institution who participates in placement drives.
Job Posting	A record describing a recruitment opportunity, including title, description, eligibility criteria, and deadline.
Eligibility	The set of criteria — school, department, programme, CGPA, backlog count, graduation year — that determine which students may apply to a given job.
Drive	A scheduled placement event organised on a particular date for a particular company.
Slot	A specific time block within an interview drive, with a defined duration and capacity.
Approved Job	A job posting that has passed both TPO assignment and faculty approval and is therefore visible to students of the assigned school(s).
RBAC	Role-Based Access Control.
OTP	One-Time Password.
SRS	Software Requirements Specification.
FR / NFR	Functional Requirement / Non-Functional Requirement.
SCIS	School of Computer and Information Sciences (representative example).
SMTP	Simple Mail Transfer Protocol.
SMS	Short Message Service.
UI	User Interface.
API	Application Programming Interface.

Term	Meaning
DB	Database.

1.4 References

- [1] IEEE/ISO/IEC 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering. Geneva: International Organization for Standardization. (Structural standard followed by this document.)
- [2] Sommerville, I. (2016). Software Engineering (10th ed.). Pearson. (Methodological reference for requirements engineering.)
- [3] Pressman, R. S., & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill. (Reference for SRS structure and quality attributes.)
- [4] WCAG 2.1 — Web Content Accessibility Guidelines. W3C Recommendation. (Accessibility benchmark referenced in §3.6.)
- [5] Institutional Placement Policy — Internal placement rules and candidate-conduct guidelines (referenced in §2.4 and §3.2 for compliance behaviours).

1.5 Overall Description

This SRS is structured into four principal sections.

Section 1 (Introduction) establishes the purpose, scope, terminology, and references.

Section 2 (Overall Description) sets the contextual frame for the system: its perspective relative to existing institutional infrastructure, a summary of its functions, the characteristics of its user classes, the constraints imposed on its operation, the assumptions on which it relies, and the prioritisation of requirements between the present release and future increments.

Section 3 (Specific Requirements) is the technical heart of the document. It enumerates interface requirements, functional requirements (with use-case-based descriptions and a data dictionary), performance and database requirements, design constraints, and the non-functional system attributes (reliability, availability, security, maintainability, portability).

Section 4 (Supporting Information) contains supplementary appendices, including a traceability matrix and notes regarding revision history.

The reader is expected to consume this document linearly. Each section assumes familiarity with the terminology established in §1.3.

2. Overall Description

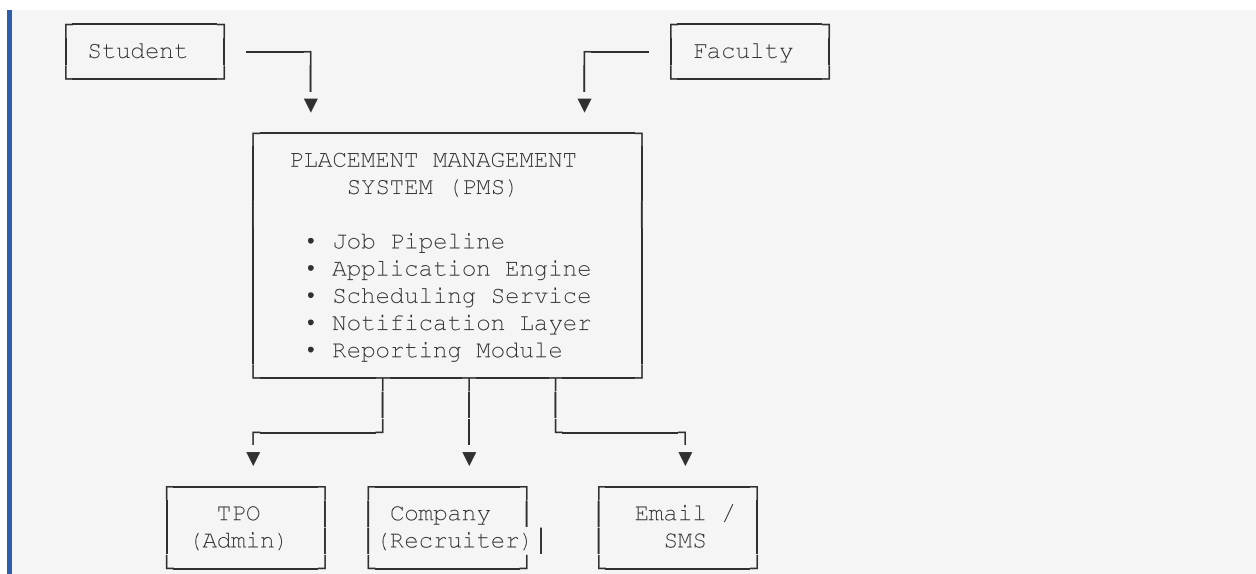
2.1 Product Perspective

The Placement Management System is a new, standalone web application developed specifically for institutional campus recruitment. It is not a derivative of, or extension to, any existing institutional information system. It is, however, designed with clear architectural boundaries so that it can later be integrated with the institution's academic records system, official email directory, and library or attendance services through well-defined application programming interfaces; such integrations are explicitly out of scope for the present release.

The system is structured along a conventional three-tier architecture:

- **Presentation Tier** — a responsive web client that provides the role-aware user interface for all four user classes.
- **Application Tier** — a stateless backend service that performs authorisation checks, business-rule validation, and orchestration of multi-step workflows (job approval cascade, exam coordination, interview slot booking).
- **Data Tier** — persistent storage of users, jobs, applications, schedules, announcements, and audit data, with referential integrity preserved at the application layer.

External interfaces include an email gateway for notifications, an optional SMS gateway for high-priority alerts, and the institution's authentication infrastructure if integrated single sign-on is later adopted. The system context is depicted below:



2.2 Product Functions

The system supports the following high-level functional capabilities. Each is decomposed in Section 3.

Capability	Description
F1 — Identity and Access Management	Registration, authentication, session management, password recovery, and role-based authorisation.
F2 — Job Lifecycle Management	Company-side creation of postings; TPO-side assignment to relevant schools; faculty-side approval or rejection; student-side discovery of approved jobs.

Capability	Description
F3 — Application Management	Per-application resume submission, eligibility-driven application acceptance, applicant data delivery to the recruiting company, candidate shortlisting.
F4 — Examination Management	Company proposal of online or offline examinations, TPO confirmation, faculty availability coordination, venue allocation, configurable test links.
F5 — Interview Management	Slot-based scheduling with time-window definition, automatic or first-come-first-served slot allocation, online meeting-link activation, and offline venue dissemination.
F6 — Announcements	Faculty-authored, role-targeted announcements with controlled visibility.
F7 — Notifications	Multi-channel (email, SMS) notification delivery for application confirmations, schedule changes, exam updates, and final results.
F8 — Reporting and Dashboards	TPO-accessible analytics on placement rates, drive counts, and operational metrics.
F9 — Account Management	User-initiated profile updates, password changes, and personal settings management.

2.3 User Characteristics

The system is designed for four distinct user classes. Each has separate role assignment, distinct operational tasks, and differentiated usability needs.

2.3.1 Student

- Volume: Several thousand active students across multiple schools.
- Technical proficiency: High; predominantly comfortable with web applications.
- Primary tasks: Discover eligible postings, apply with a tailored resume, attend exams and interviews, monitor application status, receive notifications.
- Expectations: Mobile-friendly access, transparent eligibility logic, prompt notifications, clear application-status feedback.

2.3.2 Faculty (Placement Coordinator)

- Volume: One per school.
- Technical proficiency: Moderate to high.
- Primary tasks: Review jobs assigned to their school, approve or reject them with rationale, confirm availability for offline exams, post school-level announcements.
- Expectations: Data scoped strictly to their school; clear audit trail of actions taken.

2.3.3 TPO (Training and Placement Officer)

- Volume: A single active TPO at any time.
- Technical proficiency: Moderate; functional rather than operational computer literacy.
- Primary tasks: Review newly created jobs, assign them to relevant schools, coordinate exam logistics, allocate venues, oversee interview scheduling, monitor placement metrics, generate reports.
- Expectations: A workbench surfacing pending actions first; comprehensive dashboards; export capabilities.

2.3.4 Company (Recruiter)

- Volume: Tens to low hundreds of distinct recruiter accounts per placement season.
- Technical proficiency: Variable; the interface assumes no recruitment-domain expertise.
- Primary tasks: Create job postings with eligibility criteria, view applicant data, shortlist candidates, propose exam logistics, configure interviews.
- Expectations: Clear feedback on approval status, confidential handling of applicant data, simple slot-management interface.

2.4 Constraints

The following constraints bind the design and operation of the system:

ID	Constraint	Description
C1	Web-based delivery	The system shall be accessible only through standard modern web browsers. No native mobile or desktop client is included in the present release.
C2	Role-based access control	Every operation shall be authorised against the user's role and scope; cross-role data leakage shall be impossible by design.
C3	Per-application resumes	Students shall not maintain a single global resume that is auto-attached to all applications; each application shall carry its own resume version.
C4	Approval gating	No job shall be visible to any student until both TPO assignment and faculty approval are complete.
C5	Compliance with institutional placement policy	All workflows shall enforce the institution's existing placement rules, including candidate-conduct policies and faculty-approval cascades.
C6	Standard authentication	Passwords shall be stored only as salted hashes; no plaintext credentials shall ever be persisted.
C7	Accessibility	The interface shall meet WCAG 2.1 Level AA at minimum.
C8	Notification reliability	The system shall use commercially available email and SMS gateways; the system itself shall not implement low-level mail or SMS protocols.

2.5 Assumptions and Dependencies

ID	Assumption / Dependency
A1	Each enrolled student is issued an institutional email address at the time of admission.
A2	Each school designates exactly one faculty member as the Placement Coordinator for the placement season.
A3	The institution's placement policy, including candidate-conduct rules and approval hierarchies, is communicated to the development team and reflected in the system's enforcement logic.
A4	The chosen email and SMS gateways maintain their advertised service levels; the availability targets in §3.3 assume these dependencies operate normally.
A5	Companies provide accurate eligibility criteria when creating jobs; the system enforces these criteria but does not validate their reasonableness.

ID	Assumption / Dependency
A6	Students keep their academic profile data current; stale data leading to ineligibility is the student's responsibility.
A7	Internet connectivity is available to all users at the time of system interaction; offline mode is not supported.

2.6 Apportioning Requirements

To deliver value incrementally, the requirements are partitioned between the present release (Phase 1) and future development (Phase 2+).

Present Release (Phase 1)

- Identity and access management (F1)
- Complete job lifecycle (F2)
- Per-application resume submission and shortlisting (F3)
- Online and offline examination coordination (F4)
- Slot-based interview scheduling (F5)
- Announcements (F6)
- Email and SMS notifications (F7)
- Basic dashboards and reports (F8)
- Account management (F9)

Deferred to Future Phases

- Native mobile application.
- Online proctored assessment with question banks (the present release supports only externally-linked test platforms).
- Single sign-on integration with the institution's central identity provider.
- Automatic CGPA verification through the academic records system.
- Multi-institution federation (supporting more than one institution per deployment).
- Advanced analytics with predictive insights.
- In-app video interviewing (the present release relies on third-party conferencing platforms).

3. Specific Requirements

3.1 Interface Requirements

3.1.1 External Interface

The user interface shall be a responsive single-page web application accessible from desktop and mobile browsers.

Input characteristics:

- All forms shall provide visible labels (not placeholder-only), inline validation on field blur, and explicit error states with actionable messages.
- File uploads (resumes) shall be restricted to PDF format with a per-file size limit of 5 MB.
- All date/time inputs shall use ISO 8601 internally and render in the user's local timezone.

Output characteristics:

- Listings (jobs, applicants, applications, schedules) shall support pagination with a default page size of 20 items, sorting, and filtering.
- Reports shall be exportable in CSV and PDF formats.
- Notifications shall be delivered both as in-application toasts and through email/SMS as appropriate.

Role-based UI variation:

- Each role shall have a distinct landing dashboard upon login.
- Navigation elements not permitted for the user's role shall be hidden rather than displayed and disabled.

3.1.2 Hardware Interface

The system has no direct hardware dependencies beyond the cloud or institutional servers on which it is hosted. Specifically:

- It does not interact with biometric devices, printers, or proctoring hardware.
- It does not require dedicated hardware on the client side beyond a standard web browser running on a personal computer or smartphone.
- Server-side storage shall be provisioned to accommodate at least 10 GB of resume data and metadata per active placement season.

3.1.3 Software Interface

The system shall integrate with the following external software components:

Interface	Direction	Purpose
Authentication Service	Client & Server ↔ Service	User identity verification, session token issuance, password reset workflows.
Database Management System	Server ↔ DBMS	Persistent storage of users, jobs, applications, schedules, announcements, and audit logs.
Email Gateway	Server → SMTP Provider	Delivery of OTP, approval notices, application confirmations, schedule alerts, result notifications.
SMS Gateway	Server → SMS Provider	Delivery of high-priority alerts (interview reminders, last-minute schedule changes).

Interface	Direction	Purpose
Web Browser	Client ↔ User	Rendering of the application UI; supports modern evergreen browsers (latest two major versions of Chrome, Firefox, Edge, Safari).

3.1.4 Communication Interface

CI-1. All client–server communication shall use HTTPS with TLS 1.2 or later. Plain-HTTP endpoints shall not exist in production.

CI-2. The application programming interface shall use JSON request and response bodies with Content-Type: application/json; charset=utf-8.

CI-3. Authenticated requests shall carry a session bearer token in the Authorization HTTP header.

CI-4. Cross-origin resource sharing shall be restricted to the production frontend origin and approved development origins; wildcard origin policies shall not be permitted.

CI-5. Email delivery shall use authenticated SMTP over TLS; SMS delivery shall use the chosen provider's REST API over HTTPS.

3.2 Functional Requirements

3.2.1 Use Case Model / Information Flows

The functional requirements are organised into the capability groups F1–F9 introduced in §2.2. Each capability is realised through one or more use cases, summarised below.

Capability	Primary Actors	Key Use Cases
F1 — Identity & Access	All users	Register, Sign In, Sign Out, Reset Password, Update Profile
F2 — Job Lifecycle	Company, TPO, Faculty, Student	Create Job, Assign Job to Schools, Approve/Reject Job, View Approved Jobs
F3 — Application	Student, Company	Apply with Resume, View Applicants, Shortlist Candidates
F4 — Examination	Company, TPO, Faculty, Student	Propose Exam, Confirm Schedule, Allocate Venue, Conduct Exam
F5 — Interview	Company, Student	Define Interview Window, Generate Slots, Book Slot, Attend Interview
F6 — Announcements	Faculty, all roles	Post Announcement, View Announcement
F7 — Notifications	System, all roles	Send Notification (email/SMS), Receive Notification
F8 — Reporting	TPO	View Dashboard, Generate Report, Export Report
F9 — Account	All users	Update Settings, Change Password

The principal information flow follows a controlled cascade. A company-created job enters the system in a Pending TPO Review state. The TPO reviews and assigns it to one or more schools, transitioning it to Pending Faculty Approval. Each assigned faculty independently approves or rejects within their scope. Only when all assigned faculty have approved does the job transition to

Approved and become visible to students of those schools. Applications, exams, and interviews build on this foundation in chronological order.

3.2.2 Use Case Specifications / Process Description

Selected use cases are specified below in standard form.

UC-01: Create Job (Company)

Actor: Company.

Precondition: Company is registered, approved, and signed in.

Trigger: Company submits a new job creation form.

Main Flow:

1. Company enters job title, description, location, employment type, compensation, deadline, and eligibility criteria.
2. System validates the input.
3. System persists the posting with status Pending TPO Review.
4. System notifies the TPO of the pending posting.
5. System confirms creation to the company.

Postcondition: Job exists in Pending TPO Review state.

Alternate Flow: Validation Failure — System rejects the submission with field-level error messages; no posting is created.

UC-02: Assign Job to Schools (TPO)

Actor: TPO.

Precondition: A job exists in Pending TPO Review.

Main Flow:

6. TPO opens the pending job.
7. TPO selects one or more eligible schools (e.g., SCIS, Mathematics).
8. System updates the job state to Pending Faculty Approval and records the assignment.
9. System notifies the faculty of each assigned school.

Postcondition: Job is assigned and visible to designated faculty.

UC-03: Approve or Reject Job (Faculty)

Actor: Faculty.

Precondition: A job has been assigned to the faculty's school.

Main Flow:

10. Faculty reviews the job description and eligibility.
11. Faculty selects Approve or Reject.
12. If rejected, faculty supplies a reason.
13. System records the decision and updates the job status.
14. If all assigned faculty approve, system transitions the job to Approved and notifies students of those schools.

Postcondition: Decision recorded; job status updated accordingly.

UC-04: Apply to Job (Student)

Actor: Student.

Precondition: Job is in Approved status; student belongs to an assigned school.

Main Flow:

15. Student opens the job detail view.
16. Student uploads a resume specifically tailored for this application.
17. System verifies eligibility (school, department, programme, CGPA, backlogs, graduation year).

18. System rejects duplicate applications.
 19. System creates an application record with status Applied and notifies the company.
- Postcondition:** Application record exists; company is notified.

UC-05: Shortlist Candidates (Company)

Actor: Company.

Main Flow:

20. Company opens the applicant list for one of its jobs.
21. Company reviews resumes and academic data.
22. Company marks selected applications as Shortlisted.
23. System notifies shortlisted students.

UC-06: Schedule Examination (Company → TPO → Faculty)

Actors: Company (initiator), TPO (coordinator), Faculty (consulted for offline).

Main Flow:

24. Company proposes examination type (online/offline) and candidate dates.
25. System forwards the proposal to the TPO.
26. TPO consults faculty for offline availability and confirms dates.
27. For offline exams, TPO allocates a venue.
28. System records the confirmed schedule and notifies all stakeholders.

Special Rule: Online exam links remain editable up to one hour before the scheduled start time.

UC-07: Schedule and Attend Interview (Company → Student)

Main Flow:

29. Company defines interview time window, slot duration, and mode (online/offline).
30. System generates the slot inventory.
31. Slots are either auto-assigned or made available for student first-come-first-served booking, per the company's choice.
32. Students book slots; system records bookings and confirms.
33. For online interviews, the Join Interview control becomes active five minutes before the slot start.
34. For offline interviews, venue and timing are communicated.
35. Student attends the interview.
36. Company submits the outcome.

UC-08: Post Announcement (Faculty)

Actor: Faculty.

Main Flow:

37. Faculty composes title, body, and target audience (students of own school, TPO, companies).
38. System publishes the announcement and triggers notifications.

UC-09: View Dashboard / Generate Report (TPO)

Actor: TPO.

Main Flow:

39. TPO opens the dashboard.
40. System displays placement rate, number of drives, total offers, and other key metrics.
41. TPO optionally exports the report in CSV or PDF.

3.2.3 Analysis Classes / Data Dictionary

The persistent data entities of the system are described below. The logical model is independent of the physical database technology; specific tables and indices are addressed in §3.4.

User

Attribute	Type	Description
userId	String (PK)	Primary identifier.
fullName	String	Display name.
email	String (unique)	Verified email address.
passwordHash	String	Salted password hash.
role	Enum	STUDENT FACULTY TPO COMPANY
status	Enum	ACTIVE INACTIVE SUSPENDED
createdAt	DateTime	Account creation timestamp.

Student (one-to-one with User)

Attribute	Type	Description
userId	String (FK)	References User.
rollNumber	String	Institutional roll number.
school	String	Assigned school.
department	String	Department within school.
programme	String	Degree programme.
cgpa	Float	0.0–10.0 scale.
backlogs	Integer	Active backlog count.
graduationYear	Integer	Expected year of graduation.
phoneNumber	String	Contact number.

Faculty (one-to-one with User)

Attribute	Type	Description
userId	String (FK)	References User.
school	String	The single school for which this faculty is the placement coordinator.
designation	String	Academic rank.

TPO (one-to-one with User)

Attribute	Type	Description
userId	String (FK)	References User.
office	String	TPO office identifier.

Company

Attribute	Type	Description
companyId	String (PK)	Primary identifier.
primaryContactUserId	String (FK)	The recruiter user representing this company.
companyName	String	Registered name.
website	String	URL.
industry	String	Industry classification.
description	String	Profile text.
status	Enum	PENDING APPROVED REJECTED SUSPENDED

Job

Attribute	Type	Description
jobId	String (PK)	Primary identifier.
companyId	String (FK)	Posting company.
title	String	Job title.
description	Text	Detailed description.
location	String	Work location.
employmentType	Enum	FULL_TIME INTERNSHIP PPO
ctc	Decimal	Compensation per annum.
deadline	DateTime	Application deadline.
eligibility	EligibilityRule	Embedded rule (see EligibilityRule entity).
status	Enum	PENDING_TPO ASSIGNED PENDING_FACULTY APPROVED REJECTED CLOSED
postedAt	DateTime	Creation timestamp.

EligibilityRule (embedded value object)

Attribute	Type	Description
schools	List<String>	Eligible schools.
departments	List<String>	Eligible departments.
minCgpa	Float	Minimum CGPA.
maxBacklogs	Integer	Maximum permitted backlogs.
programmes	List<String>	Eligible programmes.
graduationYears	List<Integer>	Eligible graduation years.

JobAssignment

Attribute	Type	Description
assignmentId	String (PK)	Primary identifier.
jobId	String (FK)	Job being assigned.
school	String	School to which the job is assigned.
assignedBy	String (FK)	TPO user.
assignedAt	DateTime	Timestamp.
facultyDecision	Enum	PENDING APPROVED REJECTED
decisionAt	DateTime	Faculty decision timestamp.
rejectionReason	String?	Provided when rejected.

Application

Attribute	Type	Description
applicationId	String (PK)	Primary identifier.
jobId	String (FK)	Target job.
studentId	String (FK)	Applying student.
resumePath	String	Storage path for the application-specific resume.
status	Enum	APPLIED SHORTLISTED EXAM_SCHEDULED INTERVIEWED SELECTED REJECTED WITHDRAWN
appliedAt	DateTime	Submission timestamp.

Exam

Attribute	Type	Description
examId	String (PK)	Primary identifier.
jobId	String (FK)	Associated job.
type	Enum	ONLINE OFFLINE
proposedDates	List<DateTime>	Company-proposed candidates.
confirmedDate	DateTime	TPO-confirmed final date.
venue	String?	Allocated venue (offline only).
testLink	String?	Test URL (online only).
status	Enum	PROPOSED CONFIRMED COMPLETED

Interview

Attribute	Type	Description
interviewId	String (PK)	Primary identifier.
jobId	String (FK)	Associated job.
mode	Enum	ONLINE OFFLINE
windowStart	DateTime	Earliest slot start.
windowEnd	DateTime	Latest slot end.
slotDuration	Integer	Minutes.
venue	String?	Offline venue.
meetingLink	String?	Online meeting URL.
allocationMode	Enum	AUTO FIRST_COME_FIRST_SERVED
status	Enum	SCHEDULED IN_PROGRESS COMPLETED

InterviewSlot

Attribute	Type	Description
slotId	String (PK)	Primary identifier.
interviewId	String (FK)	Parent interview.
startTime	DateTime	Slot start.
endTime	DateTime	Slot end.
applicationId	String (FK)?	Booking link.
status	Enum	OPEN BOOKED COMPLETED

Announcement

Attribute	Type	Description
announcementId	String (PK)	Primary identifier.
authorUserId	String (FK)	Faculty author.
title	String	Headline.
body	Text	Content.
visibleTo	List<Role>	Audience roles.
postedAt	DateTime	Timestamp.

Notification

Attribute	Type	Description
notificationId	String (PK)	Primary identifier.

Attribute	Type	Description
recipientUserId	String (FK)	Target user.
type	Enum	APPLICATION_CONFIRM JOB_APPROVED EXAM_SCHEDULED INTERVIEW_SCHEDULED RESULT ANNOUNCEMENT
channel	Enum	EMAIL SMS IN_APP
message	String	Body.
sentAt	DateTime	Timestamp.
isRead	Boolean	Read status (in-app).

Report

Attribute	Type	Description
reportId	String (PK)	Primary identifier.
reportType	String	Category (e.g., placement-rate).
generatedBy	String (FK)	TPO user.
generatedAt	DateTime	Timestamp.
data	JSON	Aggregated content.

3.3 Performance Requirements

3.3.1 Response Time

PR-1. Under normal load, 95% of read-API responses (job listings, application listings, dashboards) shall complete within 2 seconds at the API boundary.

PR-2. Under normal load, 95% of write-API responses (apply, post job, update status) shall complete within 3 seconds at the API boundary.

PR-3. First Contentful Paint of any authenticated page on consumer broadband (≥ 10 Mbps) shall occur within 3 seconds.

PR-4. Email and SMS notification dispatch shall begin within 30 seconds of the triggering event; delivery to the user's inbox shall normally complete within 2 minutes.

3.3.2 Throughput

PR-5. The system shall support sustained throughput of at least 100 applications submitted per minute during peak placement-season days.

PR-6. The notification subsystem shall process at least 1,000 outbound messages per hour without queue backlog.

3.3.3 Capacity

PR-7. The system shall accommodate at least 10,000 registered students, 500 registered companies, and 1,000 jobs per placement season without degradation.

PR-8. Resume uploads shall be limited to PDF format, 5 MB per file.

PR-9. A student shall maintain at most one resume per application; resume version history per application is not required.

3.3.4 Concurrency

PR-10. The system shall support at least 500 concurrent users during peak load (e.g., result publication, slot booking) without exceeding the response-time bounds in §3.3.1.

3.4 Logical Database Requirements

The system shall persist all entities defined in §3.2.3 with the following logical-level requirements:

DB-1 — Referential Integrity: Every foreign key (userId, companyId, jobId, applicationId, etc.) shall reference an existing record at write time; orphan records shall not be created.

DB-2 — Uniqueness: The User table shall enforce uniqueness of email. The JobAssignment table shall enforce uniqueness of (jobId, school). The Application table shall enforce uniqueness of (jobId, studentId).

DB-3 — Indexes: The system shall maintain indexes on: Job.status and Job.deadline (for student listings); Application.studentId and Application.jobId (for application views); Notification.recipientUserId and Notification.sentAt (for inbox listings); JobAssignment.school and JobAssignment.facultyDecision (for faculty queues).

DB-4 — Transactions: Operations that span multiple records (job approval cascade, application creation, slot booking) shall execute within database transactions to maintain consistency.

DB-5 — Backups: Database backups shall be performed at least daily and retained for at least 30 days.

DB-6 — Audit Logging: Privileged operations (job approval, account creation, role changes) shall be logged in an append-only AuditLog table containing actor identity, action type, target entity, timestamp, and source IP.

DB-7 — Retention: Placement-season records (jobs, applications, schedules) shall be retained for at least three academic years to support institutional reporting.

3.5 Design Constraints

DC-1. The system shall be implemented using a stateless backend; no session state is held on the server between requests.

DC-2. The frontend shall be delivered as a responsive single-page application from a static asset host.

DC-3. All secrets (database credentials, gateway keys, signing secrets) shall reside in environment variables or a managed secret store; no secret shall appear in source control or in the client bundle.

DC-4. Source control shall use Git, with mandatory code review on every merge to the main branch.

DC-5. The application programming interface shall follow a noun-based URL scheme (e.g., /api/jobs, /api/applications/{id}); HTTP verbs shall convey the action.

DC-6. The system shall include automated unit tests for authentication, authorisation, and eligibility logic, with a minimum line coverage of 70% on these modules.

DC-7. All user-facing date and time values shall display in the user's local timezone; internal storage shall be in UTC.

3.6 Software System Attributes

3.6.1 Reliability

R-1. The system shall not lose any successfully acknowledged write under any single-instance failure; database durability guarantees shall be relied upon for this property.

R-2. Following a failed deployment, the system shall roll back to the previous known-good version automatically within 5 minutes.

R-3. All server errors (HTTP 5xx) shall be centrally logged. An error rate exceeding 1% over a 15-minute window shall trigger an automated alert.

R-4. Critical workflows (apply, approve, book slot) shall be idempotent at the API level so that a retried request following a network failure does not produce duplicate records.

3.6.2 Availability

A-1. The system shall target 99.5% monthly uptime during placement seasons (typically August through May), excluding scheduled maintenance windows announced at least 48 hours in advance.

A-2. Off-season target uptime shall be 99.0%.

A-3. Rolling deployments shall be used; no single deployment shall exceed 60 seconds of user-visible downtime.

A-4. Scheduled maintenance shall be performed outside core operational hours wherever possible.

3.6.3 Security

S-1. All client–server traffic shall be encrypted with TLS 1.2 or later.

S-2. Passwords shall be stored only as salted hashes using a modern key-derivation function (bcrypt, Argon2, or scrypt).

S-3. Strict role-based access control shall be enforced on every protected endpoint. Faculty shall see data only for their assigned school; students shall see only approved jobs for their school; companies shall see only their own applicants; the TPO shall have oversight scoped to administrative tasks.

S-4. The system shall apply request rate limiting: 20 requests per minute per IP for unauthenticated endpoints and 100 requests per minute per IP for authenticated endpoints.

S-5. Privileged actions (job approval, account state changes, role assignment) shall be recorded in the audit log per DB-6.

S-6. The system shall set standard HTTP security headers (Content-Security-Policy, Strict-Transport-Security, X-Content-Type-Options, X-Frame-Options).

S-7. File uploads shall be scanned for permitted MIME types (PDF only for resumes) and shall reject content whose magic bytes do not match the declared type.

S-8. The system shall comply with the institution's data-protection policies, including the right of users to request export and erasure of their personal data via the TPO.

3.6.4 Maintainability

M-1. The codebase shall be modular: presentation, application, and data-access layers shall be independently testable.

M-2. All API endpoints shall be documented in OpenAPI 3.0 format, with the specification versioned alongside the source code.

M-3. Each request shall produce a structured JSON log line containing method, path, user identifier, role, status code, and latency.

M-4. Configuration shall be externalised; rebuilding the application shall not be required to change connection strings, gateway keys, or feature flags.

M-5. Code review shall be mandatory; no change shall be merged to the main branch without at least one peer approval.

3.6.5 Portability

- P-1.** The backend shall be deployable as a container image and shall not rely on host-specific binaries; redeployment from one cloud provider to another shall not require code changes.
- P-2.** The frontend shall build to a static asset bundle deployable on any static hosting service.
- P-3.** The database access layer shall use a standard interface that permits migration between supported relational and document database providers with localised changes only.
- P-4.** The system shall function on the latest two major versions of Chrome, Firefox, Edge, and Safari.

4. Supporting Information

4.1 Traceability

Every functional requirement in this document carries an identifier in the form F<n> or UC-<nn>. Every non-functional requirement carries an identifier from the namespace PR-, DB-, DC-, R-, A-, S-, M-, or P-. These identifiers shall appear in:

- The Design Document, against the components that realise them.
- The Test Plan, against at least one test case per requirement.
- The User Acceptance script, against at least one acceptance check per externally observable requirement.

A consolidated traceability matrix is maintained as a separate appendix in the Test Plan.

4.2 Process Model

The system shall be developed using an iterative process organised into sprints. The development process shall include sprint planning, daily stand-ups, sprint review and demonstration, and a retrospective. Scope changes during a sprint shall be restricted to the Product Backlog and shall not be applied to the in-flight Sprint Backlog.

4.3 Open Issues

- The choice of email and SMS gateway providers is to be finalised based on cost negotiation; this SRS does not bind to a specific provider.
- Single sign-on integration with the institutional identity provider is deferred pending an inter-departmental agreement.
- The exact academic-year reporting format required by accreditation bodies shall be confirmed before the reporting module enters acceptance testing.

4.4 Glossary of Process Terms

Term	Meaning in this System
Drive	A scheduled placement event organised on a particular date for a particular company.
Placement Season	The academic-year window during which the majority of hiring activity takes place.
Eligibility Cascade	The sequence of TPO assignment, faculty approval, and student-side filtering that determines whether a job is visible to a given student.
Slot Allocation Mode	The mechanism by which interview slots are bound to students — automatic or first-come-first-served.
Approval Quorum	The condition under which a job becomes Approved — namely, that all assigned faculty have approved it.

4.5 Approval and Baselining

This Software Requirements Specification is baselined as Version 1.0 on the date of acceptance by the project supervisor. Subsequent revisions shall be tracked in a revision-history appendix and shall require re-review by the supervisor before implementation begins. Material changes — to the

role model, the workflow cascade, the policy enforcement, or the chosen technology stack — shall require explicit approval and re-baselining.

4.6 Document End

This document ends here. Subsequent artefacts — the Design Document, the Test Plan, the Test Report, and the Project Management plan — elaborate on, and are bound by, the requirements specified herein.

— END OF DOCUMENT —
